

Pyridine Phosphination - 3 categories - Model Testing (10% holdout)

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Model Search

formula	McFadden R2
class ~ para_P + NPA_6_SM + fr_aryl_methyl	0.638
class ~ NPA_1_P + loc.B5_1_SM + fr_bicyclic	0.616
class ~ Dist_.P.C._P + NPA_6_SM + para_SM	0.607
class ~ NPA_6_SM + para_SM + fr_aryl_methyl	0.607
class ~ NPA_5_P + Dist_.P.C._P + fr_aryl_methyl	0.601
class ~ NPA_1_P + Total_P + fr_bicyclic	0.597
class ~ Dist_.P.C._P + loc.B1_2_SM + para.angle_SM	0.573
class ~ para_P + NPA_6_SM + NPA_sum_SM	0.569
class ~ NPA_6_SM + NPA_sum_SM + para_SM	0.568
class ~ NPA_1_P + dip_y_P + fr_bicyclic	0.563

Section Results Summary

Model	Training Accuracy	Test Accuracy
1. 1st Place	90.62	66.67
2. 2nd Place	71.88	66.67
3. 3rd Place	84.38	100.00
4. 4th Place	87.50	77.78
5. 5th Place	71.88	77.78
6. 6th Place	65.62	77.78
7. 7th Place	84.38	88.89
8. 8th Place	78.12	100.00
9. 9th Place	84.38	100.00
10. 10th Place	62.50	77.78

1st Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
90.62%	0.631

Model Coefficients

Term	Estimate
:-----:	-----:
para_P	0.1203336
NPA_6_SM	49.8851609
fr_aryl_methyl	-4.8634393
(Intercept)	195.5366760

Training Set

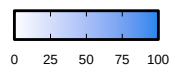
1	5 (71%)	2 (29%)	0 (0%)	7 (22%)
2	0 (0%)	18 (100%)	0 (0%)	18 (56%)
3	0 (0%)	1 (14%)	6 (86%)	7 (22%)
total	5 (100%)	21 (86%)	6 (100%)	32 (91%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	100	0	0	1
AF	100	0	0	1
B	21	79	0	1
C	59	41	0	1
D	44	56	0	1
E	100	0	0	1
F	85	15	0	1
AA	1	94	4	2
AB	42	58	0	2
AC	40	60	0	2
AE	0	50	50	2
AI	19	80	0	2
AK	1	91	9	2
AL	10	89	1	2
H	12	87	0	2
I	1	94	5	2
J	1	94	5	2
L	0	75	25	2
M	1	89	10	2
S	0	78	22	2
T	1	94	4	2
V	39	61	0	2
W	0	52	48	2
Y	4	95	1	2
Z	18	82	0	2
AN	0	39	61	3
AP	0	5	95	3
AQ	1	93	7	3
AR	0	2	98	3
AS	0	2	98	3
AT	0	5	95	3
AU	0	44	56	3



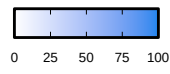
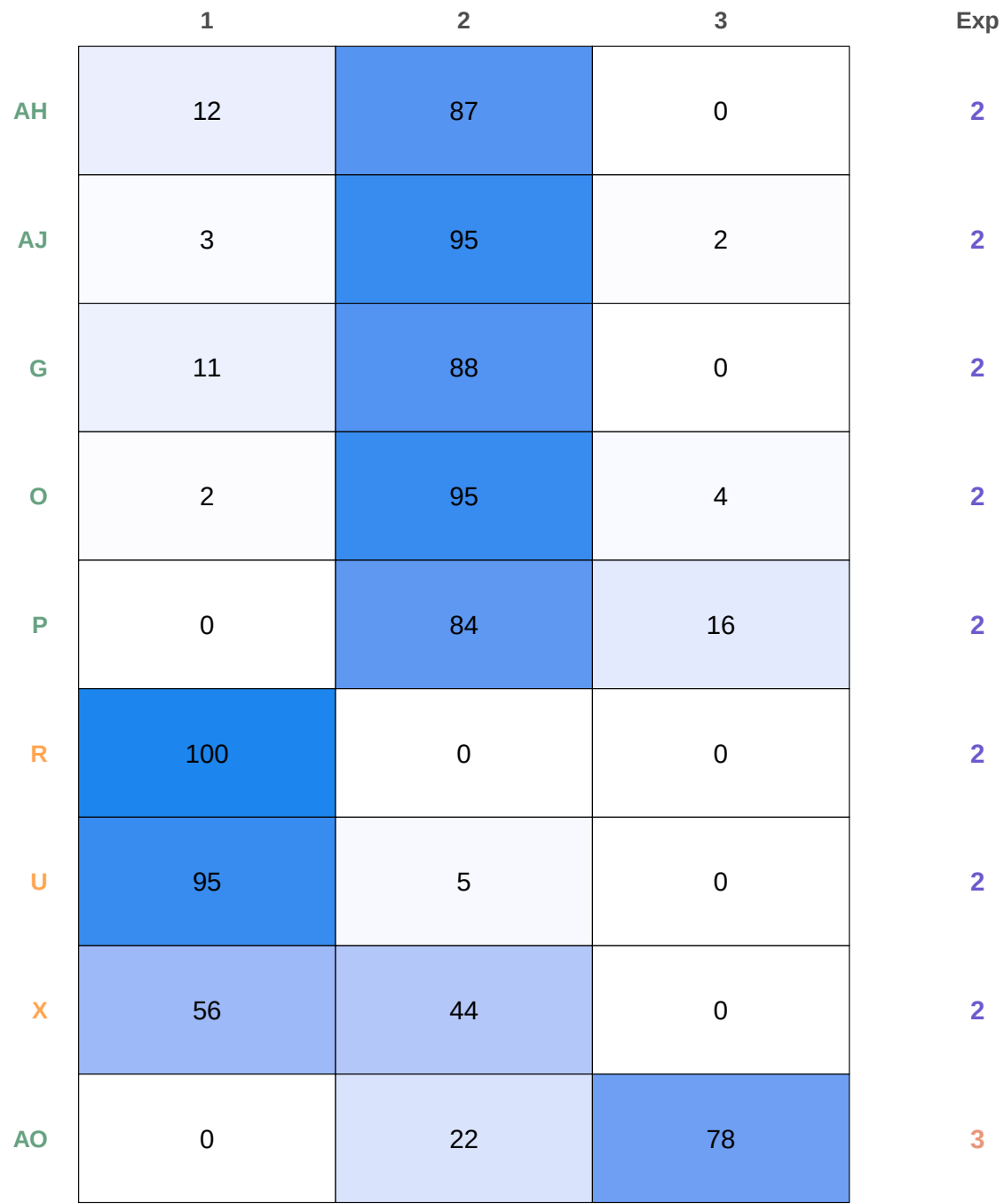
1. 1st Place

Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	3 (38%)	5 (62%)	0 (0%)	8 (89%)
3	0 (0%)	0 (0%)	1 (100%)	1 (11%)
total	3 (0%)	5 (100%)	1 (100%)	9 (67%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap



2nd Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
71.88%	0.382

Model Coefficients

Term	Estimate
:-----	-----:
NPA_1_P	66.2982853
loc.B5_1_SM	-0.6024916
fr_bicyclic	-0.3863870
(Intercept)	52.3791826

Training Set

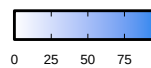
1	3 (43%)	4 (57%)	0 (0%)	7 (22%)
2	0 (0%)	17 (94%)	1 (6%)	18 (56%)
3	0 (0%)	4 (57%)	3 (43%)	7 (22%)
total	3 (100%)	25 (68%)	4 (75%)	32 (72%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	97	3	0	1
AF	25	72	3	1
B	48	51	1	1
C	78	22	0	1
D	89	11	0	1
E	13	80	7	1
F	45	54	1	1
AA	21	75	4	2
AB	8	81	12	2
AC	25	72	3	2
AE	46	53	1	2
AI	2	66	32	2
AK	5	78	17	2
AL	8	81	12	2
H	2	66	32	2
I	22	74	4	2
J	20	76	4	2
L	3	72	25	2
M	15	79	6	2
S	17	78	5	2
T	41	57	2	2
V	21	75	4	2
W	19	77	5	2
Y	0	2	98	2
Z	9	81	10	2
AN	2	65	33	3
AP	4	74	22	3
AQ	0	3	97	3
AR	0	16	84	3
AS	3	70	27	3
AT	0	7	93	3
AU	1	50	49	3



2. 2nd Place

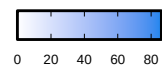
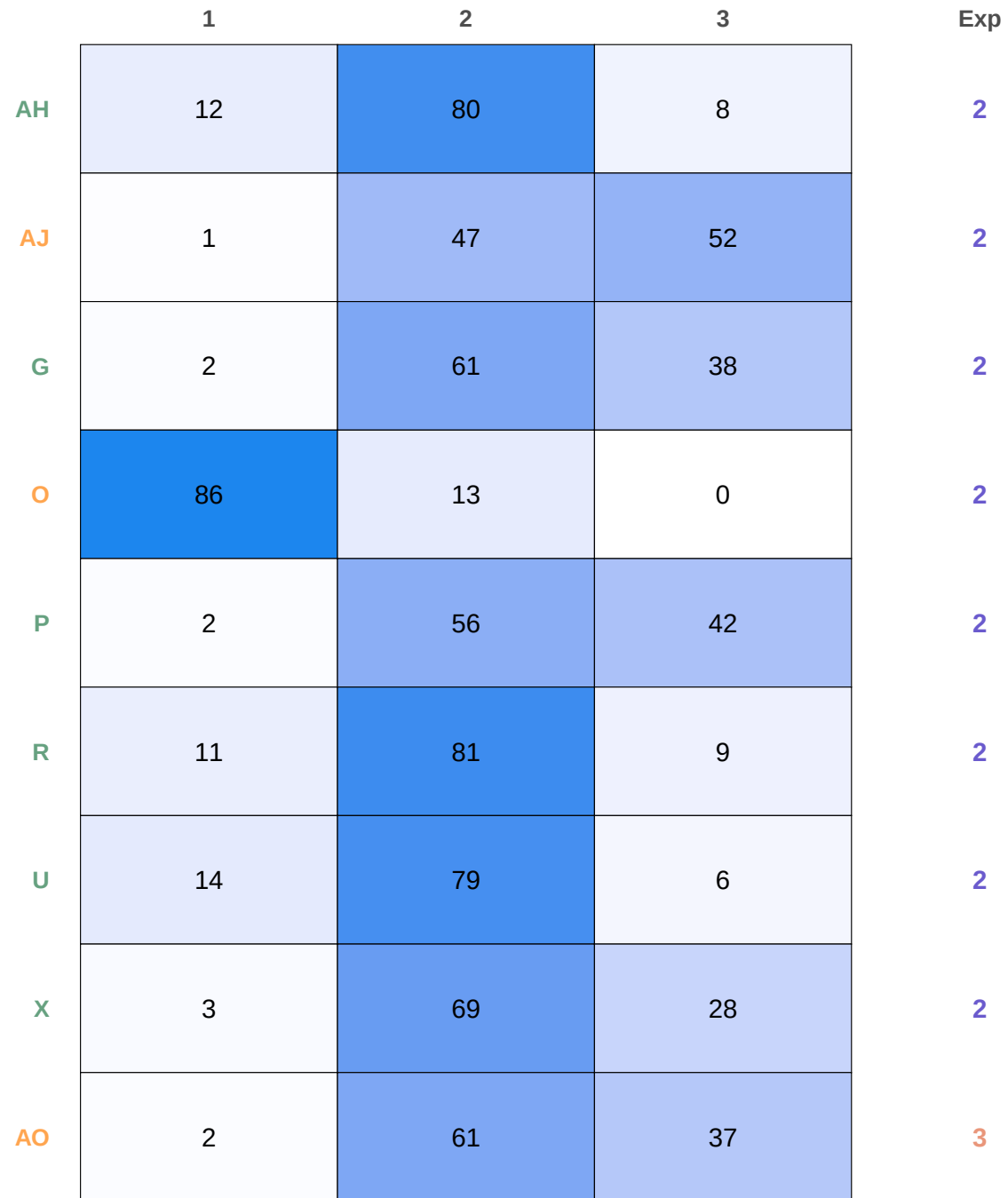
Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	1 (12%)	6 (75%)	1 (12%)	8 (89%)
3	0 (0%)	1 (100%)	0 (0%)	1 (11%)
total	1 (0%)	7 (86%)	1 (0%)	9 (67%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

2. 2nd Place

Test Set
Probability Heatmap



3rd Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
84.38%	0.607

Model Coefficients

Term	Estimate
:-----:	-----:
Dist_.P.C._P	-129.9237128
NPA_6_SM	28.2302879
para_SM	0.1585847
(Intercept)	29.5883197

Training Set

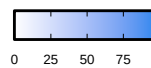
1	5 (71%)	2 (29%)	0 (0%)	7 (22%)
2	1 (6%)	16 (89%)	1 (6%)	18 (56%)
3	0 (0%)	1 (14%)	6 (86%)	7 (22%)
total	6 (83%)	19 (84%)	7 (86%)	32 (84%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	40	59	0	1
AF	78	22	0	1
B	74	26	0	1
C	50	49	0	1
D	96	4	0	1
E	41	59	0	1
F	99	1	0	1
AA	20	80	0	2
AB	31	69	0	2
AC	18	82	0	2
AE	1	92	7	2
AI	2	95	3	2
AK	0	19	81	2
AL	2	95	3	2
H	25	75	0	2
I	15	85	0	2
J	2	95	3	2
L	1	90	9	2
M	0	86	14	2
S	58	42	0	2
T	0	83	17	2
V	40	60	0	2
W	0	75	25	2
Y	3	95	3	2
Z	2	95	4	2
AN	0	13	87	3
AP	0	4	96	3
AQ	2	94	4	3
AR	0	9	91	3
AS	0	25	75	3
AT	0	16	84	3
AU	0	8	92	3

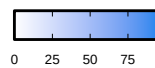
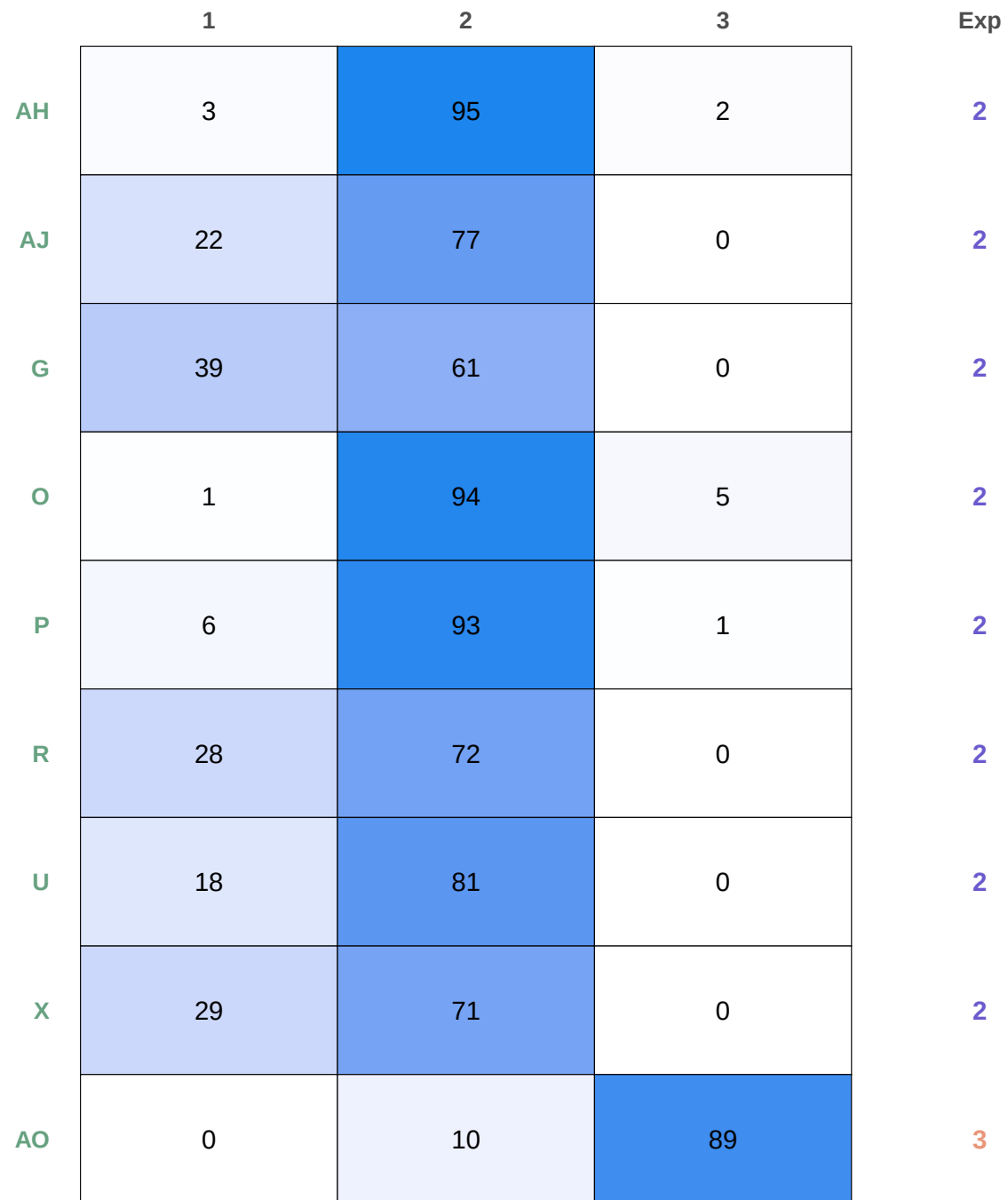


Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	0 (0%)	8 (100%)	0 (0%)	8 (89%)
3	0 (0%)	0 (0%)	1 (100%)	1 (11%)
total	0 (0%)	8 (100%)	1 (100%)	9 (100%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap



4th Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
87.5%	0.576

Model Coefficients

Term	Estimate
:-----:	-----:
NPA_6_SM	26.9990074
para_SM	0.0873063
fr_aryl_methyl	-2.9287943
(Intercept)	143.7099307

Training Set

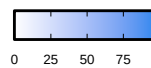
1	4 (57%)	3 (43%)	0 (0%)	7 (22%)
2	0 (0%)	18 (100%)	0 (0%)	18 (56%)
3	0 (0%)	1 (14%)	6 (86%)	7 (22%)
total	4 (100%)	22 (82%)	6 (100%)	32 (88%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	99	1	0	1
AF	97	3	0	1
B	31	67	1	1
C	8	85	7	1
D	27	72	2	1
E	99	1	0	1
F	82	18	0	1
AA	7	86	8	2
AB	31	68	1	2
AC	29	70	1	2
AE	2	74	25	2
AI	24	74	2	2
AK	3	80	18	2
AL	18	79	3	2
H	22	76	2	2
I	5	85	11	2
J	19	78	2	2
L	2	78	20	2
M	3	81	16	2
S	5	84	11	2
T	4	82	14	2
V	34	65	1	2
W	2	71	27	2
Y	7	86	7	2
Z	27	71	2	2
AN	0	17	83	3
AP	0	8	92	3
AQ	6	85	9	3
AR	0	6	94	3
AS	0	18	82	3
AT	0	27	72	3
AU	0	14	86	3

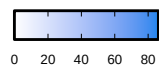
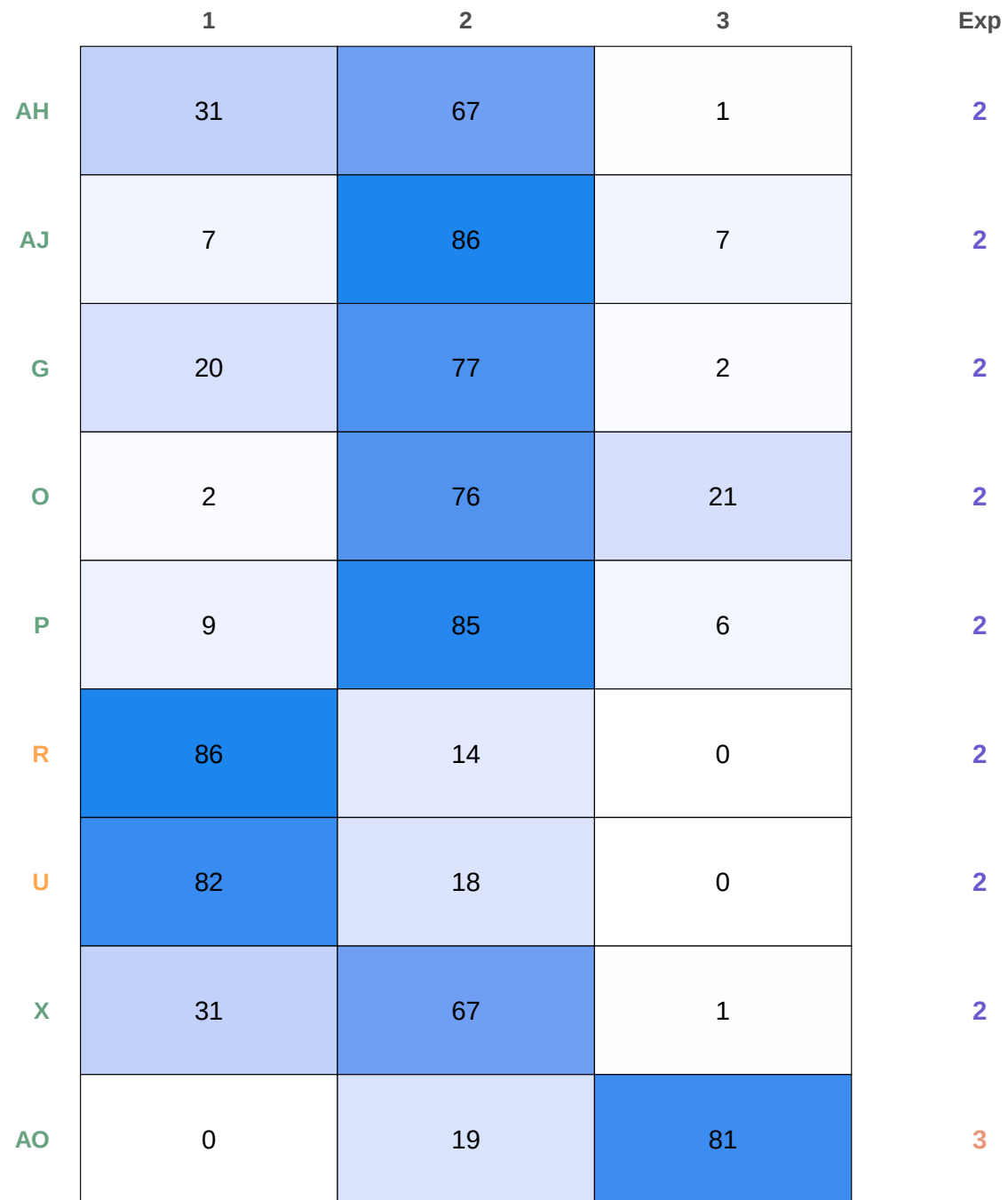


Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	2 (25%)	6 (75%)	0 (0%)	8 (89%)
3	0 (0%)	0 (0%)	1 (100%)	1 (11%)
total	2 (0%)	6 (100%)	1 (100%)	9 (78%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap



5th Place

Table:
Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
71.88%	0.446

Model Coefficients

Term	Estimate
:-----:	-----:
NPA_5_P	-57.766540
Dist_.P.C._P	-126.982301
fr_aryl_methyl	-2.492589
(Intercept)	-208.866819

Training Set

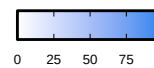
1	2 (29%)	5 (71%)	0 (0%)	7 (22%)
2	2 (11%)	16 (89%)	0 (0%)	18 (56%)
3	0 (0%)	2 (29%)	5 (71%)	7 (22%)
total	4 (50%)	23 (70%)	5 (100%)	32 (72%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	45	53	2	1
AF	78	21	0	1
B	28	68	4	1
C	20	74	7	1
D	33	63	3	1
E	99	1	0	1
F	38	59	3	1
AA	25	70	5	2
AB	5	70	24	2
AC	3	57	41	2
AE	7	73	20	2
AI	3	57	40	2
AK	22	72	6	2
AL	35	62	3	2
H	51	47	2	2
I	5	68	27	2
J	17	75	8	2
L	8	75	18	2
M	5	68	27	2
S	19	74	7	2
T	17	75	8	2
V	9	75	16	2
W	25	70	5	2
Y	56	43	1	2
Z	11	76	13	2
AN	1	43	56	3
AP	0	21	79	3
AQ	2	48	51	3
AR	2	50	48	3
AS	2	45	54	3
AT	2	56	42	3
AU	1	33	66	3

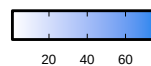
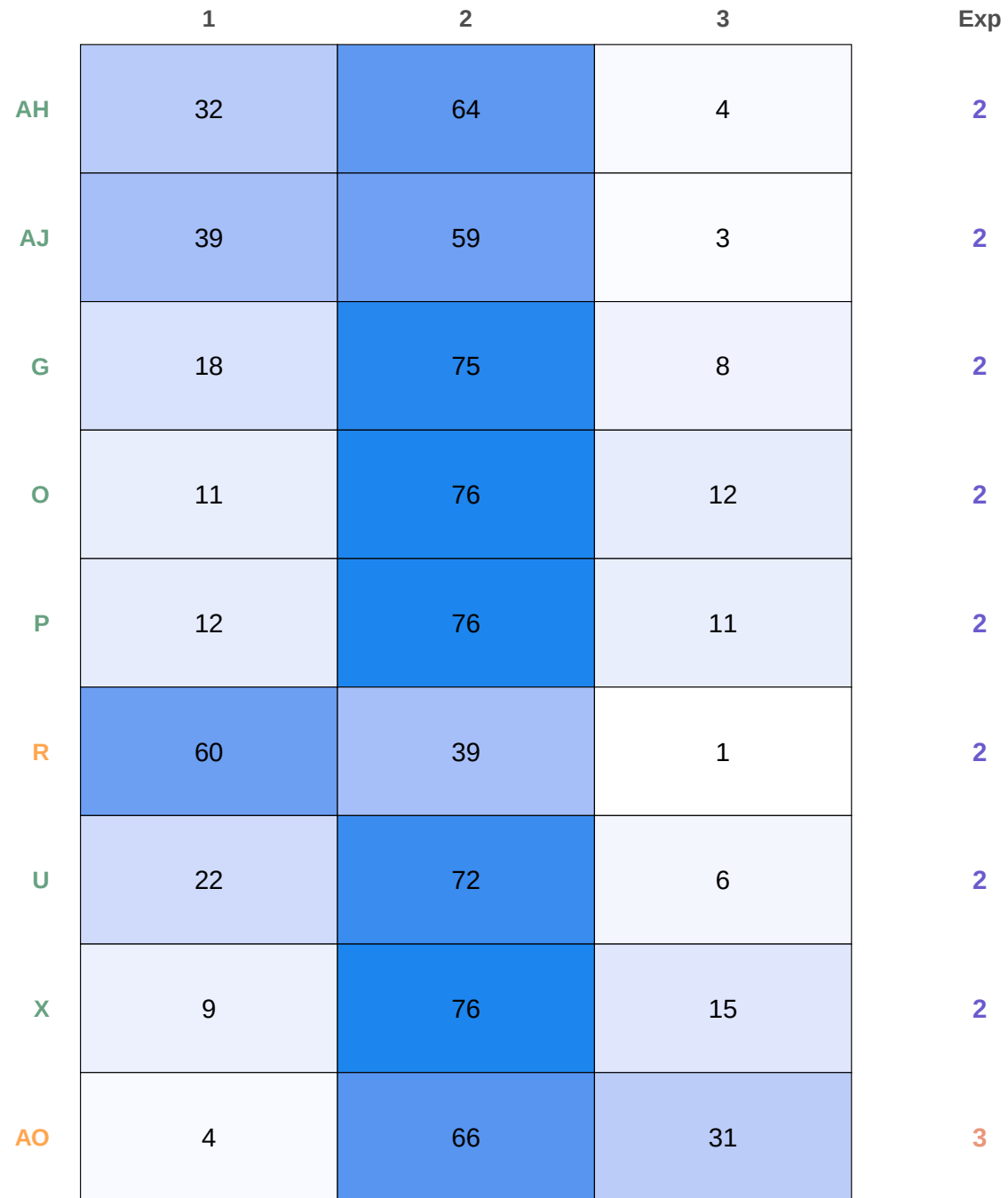


Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	1 (12%)	7 (88%)	0 (0%)	8 (89%)
3	0 (0%)	1 (100%)	0 (0%)	1 (11%)
total	1 (0%)	8 (88%)	0 (0%)	9 (78%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap



6th Place

Table:
Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
65.62%	0.363

Model Coefficients

Term	Estimate
:-----	-----:
NPA_1_P	52.9273956
Total_P	0.3280758
fr_bicyclic	-0.5910344
(Intercept)	45.8741923

Training Set

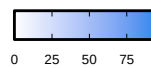
1	3 (43%)	4 (57%)	0 (0%)	7 (22%)
2	2 (11%)	14 (78%)	2 (11%)	18 (56%)
3	0 (0%)	3 (43%)	4 (57%)	7 (22%)
total	5 (60%)	21 (67%)	6 (67%)	32 (66%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	96	4	0	1
AF	32	66	2	1
B	55	44	1	1
C	68	32	0	1
D	24	74	3	1
E	15	81	5	1
F	48	51	1	1
AA	23	75	3	2
AB	1	56	43	2
AC	9	83	8	2
AE	61	38	1	2
AI	7	83	10	2
AK	20	77	3	2
AL	11	83	6	2
H	6	82	12	2
I	12	82	6	2
J	39	60	1	2
L	4	80	16	2
M	12	82	5	2
S	41	58	1	2
T	84	16	0	2
V	17	79	4	2
W	1	54	45	2
Y	0	26	74	2
Z	1	44	55	2
AN	0	19	81	3
AP	1	60	39	3
AQ	1	43	56	3
AR	1	52	47	3
AS	1	44	55	3
AT	0	10	90	3
AU	2	72	26	3

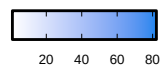
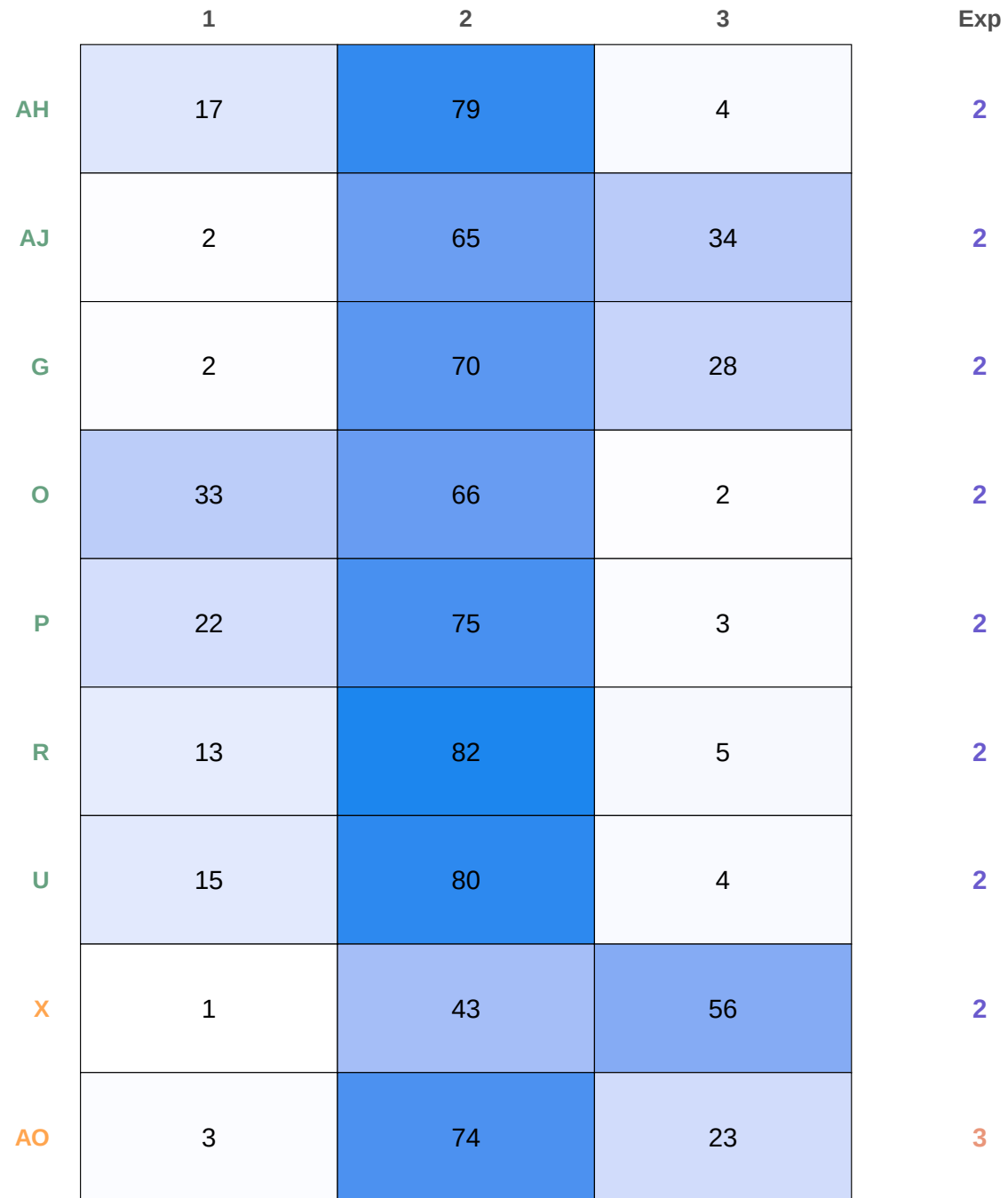


Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	0 (0%)	7 (88%)	1 (12%)	8 (89%)
3	0 (0%)	1 (100%)	0 (0%)	1 (11%)
total	0 (0%)	8 (88%)	1 (0%)	9 (78%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap



7th Place

Table:
Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
84.38%	0.573

Model Coefficients

Term	Estimate
:-----:	-----:
Dist_.P.C._P	-214.0573631
loc.B1_2_SM	0.7163644
para.angle_SM	0.1594160
(Intercept)	-380.2957090

Training Set

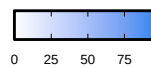
1	5 (71%)	2 (29%)	0 (0%)	7 (22%)
2	1 (6%)	16 (89%)	1 (6%)	18 (56%)
3	0 (0%)	1 (14%)	6 (86%)	7 (22%)
total	6 (83%)	19 (84%)	7 (86%)	32 (84%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	98	2	0	1
AF	4	92	4	1
B	70	30	0	1
C	49	51	0	1
D	97	3	0	1
E	97	3	0	1
F	98	2	0	1
AA	3	91	5	2
AB	3	91	6	2
AC	4	92	4	2
AE	12	86	1	2
AI	0	26	74	2
AK	7	90	2	2
AL	1	80	19	2
H	70	30	0	2
I	1	87	11	2
J	24	75	1	2
L	1	87	12	2
M	0	71	28	2
S	17	82	1	2
T	7	90	2	2
V	1	82	18	2
W	26	74	1	2
Y	1	87	11	2
Z	3	91	6	2
AN	0	30	70	3
AP	0	16	84	3
AQ	3	91	7	3
AR	0	28	72	3
AS	0	12	88	3
AT	0	10	90	3
AU	0	16	84	3

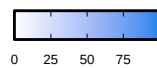
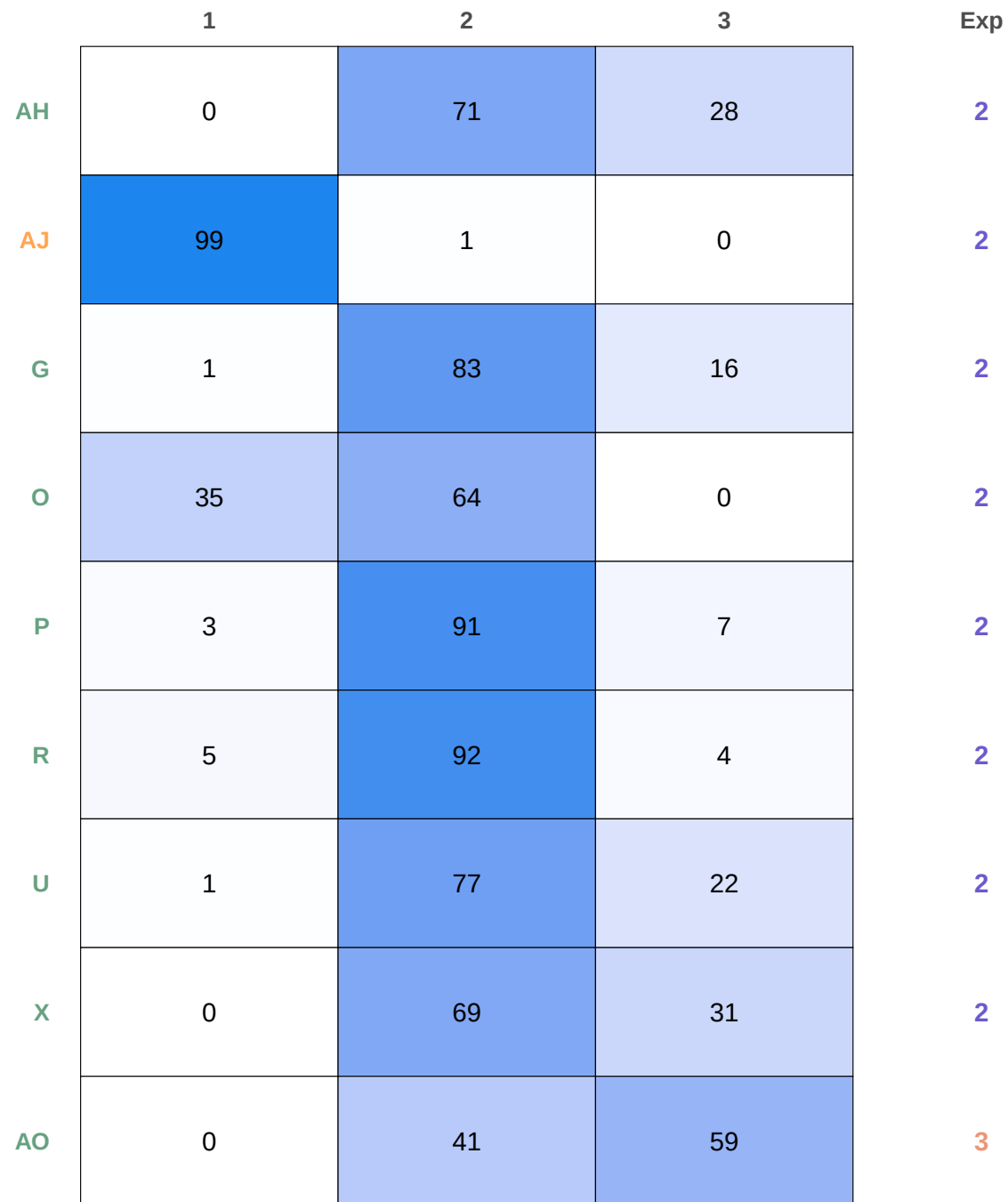


Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	1 (12%)	7 (88%)	0 (0%)	8 (89%)
3	0 (0%)	0 (0%)	1 (100%)	1 (11%)
total	1 (0%)	7 (100%)	1 (100%)	9 (89%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap



8th Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
78.12%	0.569

Model Coefficients

Term	Estimate
:-----:	-----:
para_P	0.0675549
NPA_6_SM	29.3619463
NPA_sum_SM	-14.2129597
(Intercept)	119.6618976

Training Set

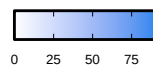
1	4 (57%)	3 (43%)	0 (0%)	7 (22%)
2	2 (11%)	15 (83%)	1 (6%)	18 (56%)
3	0 (0%)	1 (14%)	6 (86%)	7 (22%)
total	6 (67%)	19 (79%)	7 (86%)	32 (78%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	23	76	1	1
AF	58	42	0	1
B	45	55	0	1
C	80	20	0	1
D	78	22	0	1
E	35	65	0	1
F	78	22	0	1
AA	0	75	25	2
AB	8	90	2	2
AC	7	91	2	2
AE	18	81	1	2
AI	67	33	0	2
AK	0	72	28	2
AL	3	92	6	2
H	28	71	0	2
I	7	91	2	2
J	5	92	3	2
L	2	91	6	2
M	4	92	4	2
S	38	62	0	2
T	71	29	0	2
V	9	89	2	2
W	0	27	73	2
Y	1	87	11	2
Z	3	92	5	2
AN	0	40	60	3
AP	0	9	91	3
AQ	0	74	25	3
AR	0	9	91	3
AS	0	9	91	3
AT	0	16	84	3
AU	0	38	62	3

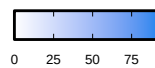
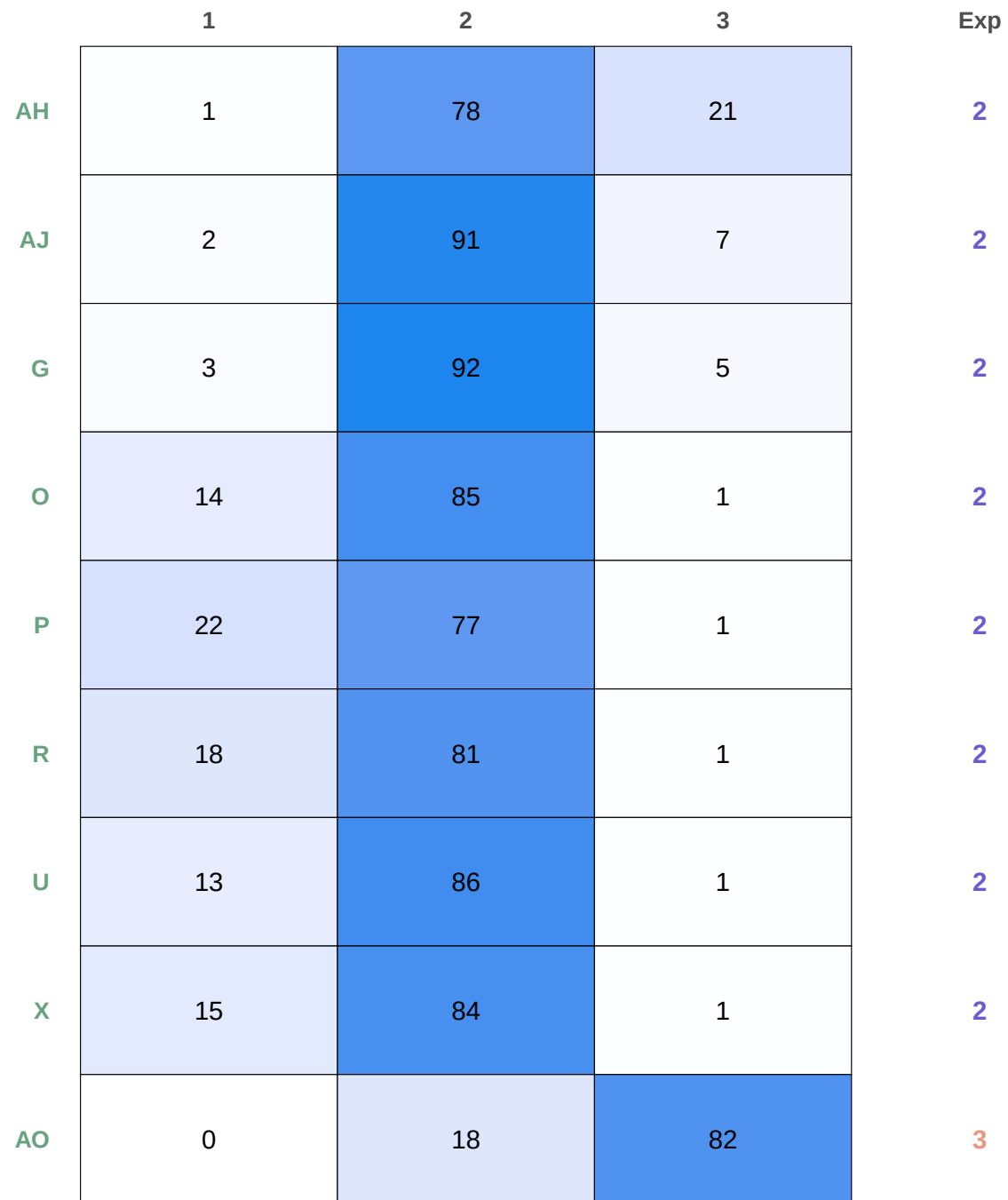


Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	0 (0%)	8 (100%)	0 (0%)	8 (89%)
3	0 (0%)	0 (0%)	1 (100%)	1 (11%)
total	0 (0%)	8 (100%)	1 (100%)	9 (100%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap



9th Place

Table:
Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
84.38%	0.568

Model Coefficients

Term	Estimate
:-----:-----:	
NPA_6_SM	31.3033671
NPA_sum_SM	-9.5625439
para_SM	0.0986009
(Intercept)	169.8906789

Training Set

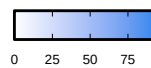
1	5 (71%)	2 (29%)	0 (0%)	7 (22%)
2	1 (6%)	16 (89%)	1 (6%)	18 (56%)
3	0 (0%)	1 (14%)	6 (86%)	7 (22%)
total	6 (83%)	19 (84%)	7 (86%)	32 (84%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	50	50	0	1
AF	74	26	0	1
B	61	39	0	1
C	27	73	0	1
D	68	32	0	1
E	42	57	0	1
F	95	5	0	1
AA	2	91	7	2
AB	18	81	1	2
AC	16	84	1	2
AE	15	85	1	2
AI	66	34	0	2
AK	0	57	43	2
AL	2	92	6	2
H	41	59	0	2
I	8	91	2	2
J	8	90	2	2
L	3	93	4	2
M	4	93	3	2
S	40	60	0	2
T	36	63	0	2
V	23	76	0	2
W	0	30	69	2
Y	1	80	19	2
Z	3	93	4	2
AN	0	16	84	3
AP	0	5	95	3
AQ	2	91	7	3
AR	0	5	95	3
AS	0	18	82	3
AT	0	29	71	3
AU	0	12	88	3

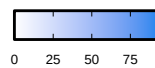
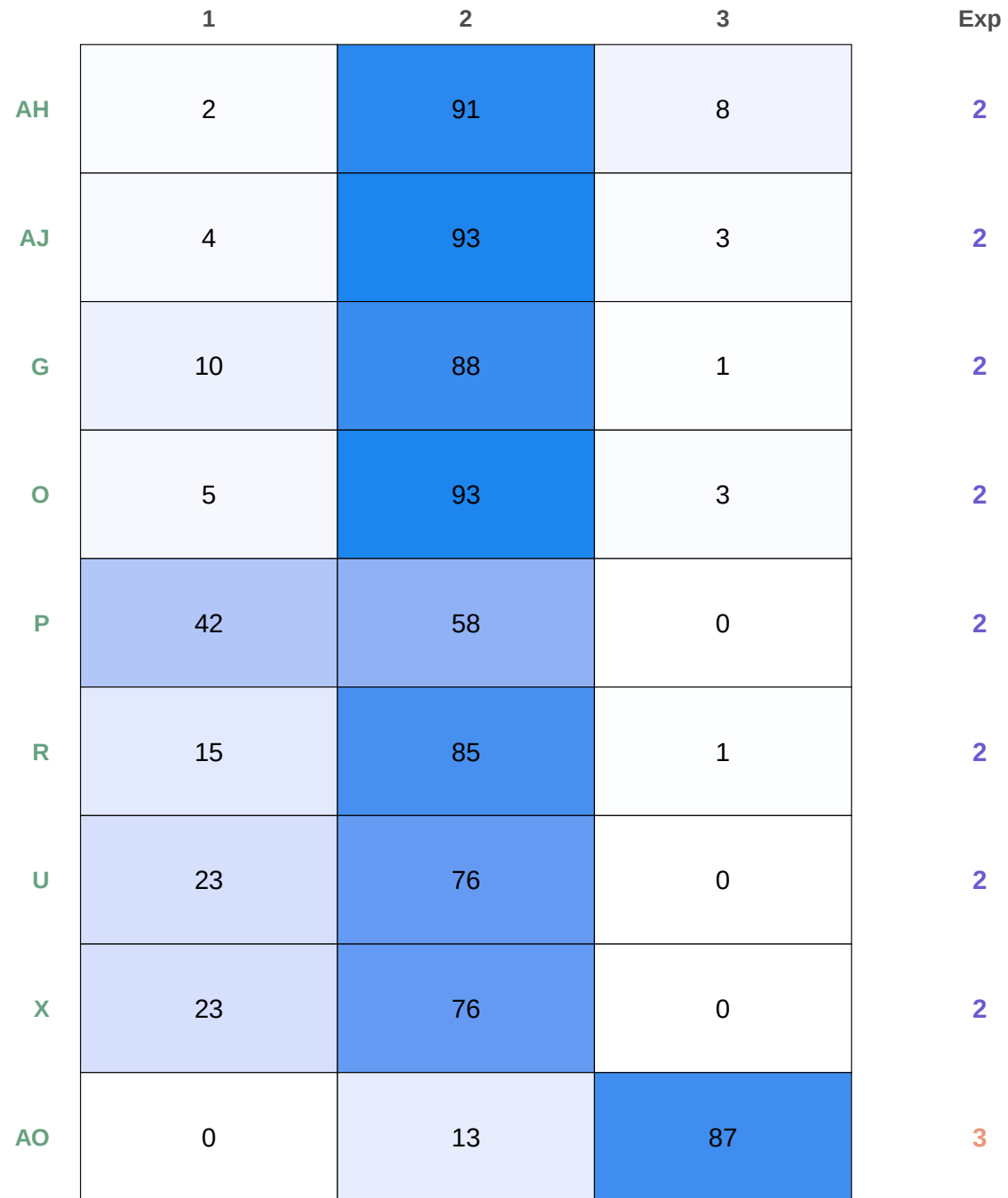


Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	0 (0%)	8 (100%)	0 (0%)	8 (89%)
3	0 (0%)	0 (0%)	1 (100%)	1 (11%)
total	0 (0%)	8 (100%)	1 (100%)	9 (100%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap



10th Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
62.5%	0.321

Model Coefficients

Term	Estimate
:-----	-----:
NPA_1_P	53.8825389
dip_y_P	0.2395465
fr_bicyclic	0.0311549
(Intercept)	45.2286400

Training Set

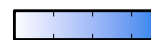
1	4 (57%)	3 (43%)	0 (0%)	7 (22%)
2	3 (17%)	13 (72%)	2 (11%)	18 (56%)
3	0 (0%)	4 (57%)	3 (43%)	7 (22%)
total	7 (57%)	20 (65%)	5 (60%)	32 (62%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	3	Exp
A	92	8	0	1
AF	24	73	4	1
B	32	65	2	1
C	55	44	1	1
D	76	24	0	1
E	19	76	5	1
F	58	41	1	1
AA	17	77	5	2
AB	3	71	26	2
AC	10	80	10	2
AE	63	36	1	2
AI	4	74	22	2
AK	6	79	15	2
AL	19	76	5	2
H	5	77	17	2
I	6	78	15	2
J	62	38	1	2
L	4	74	22	2
M	12	80	8	2
S	17	77	6	2
T	78	22	0	2
V	11	80	9	2
W	3	67	30	2
Y	0	12	88	2
Z	1	35	64	2
AN	2	64	34	3
AP	3	66	32	3
AQ	1	37	63	3
AR	2	56	42	3
AS	1	44	55	3
AT	0	28	71	3
AU	2	62	36	3



0 25 50 75

Test Set

1	0 (0%)	0 (0%)	0 (0%)	0 (0%)
2	1 (12%)	7 (88%)	0 (0%)	8 (89%)
3	0 (0%)	1 (100%)	0 (0%)	1 (11%)
total	1 (0%)	8 (88%)	0 (0%)	9 (78%)
	1	2	3	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap

